Machine Learning Solutions for Societal Problems: Context sensitive retrieval of information from documents

PGDip Applied Data Science

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Continuous Assessment (CA1)

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Abstract

This paper explores the concept of a machine learning-based solution to enable a person to ask a question, and receive a natural language response based on the context of the question, where the answer to the question is expected to be derived from a specific knowledge-base database compiled from an organisation's internal documents.

1 Problem Identification

Problem Context

All organisations are managed according to a set of rules (1). These establish the purpose, objectives and strategies of the organisation, define how it should operate, and how it takes decisions. This is the subject of *governance*, and it is important for many reasons, including:

1. *Shareholders:* the owners of the organisation, who typically make financial investment expect it to succeed based on clear objectives in line with expectations;
2. *Performance:* a future vision and organisational objectives are performance targets that management are accountable to deliver;
3. *Legal & regulatory compliance:* many organisations operate within legal and regulatory constraints and have mandated obligations to fulfill;
4. *Risk management:* ensuring success requires mitigating the chance of failure and preventing problems that impede desired results;
5. *Stakeholder confidence:* organisations interact with many types of stakeholders, and the culture, brand and reputation of an organisation enables attracting, retaining and better engagement with all stakeholders.

An organisation's *rulebook* is typically formalised through a set of internal documents that establish the *policies and procedures* for all materially important activities. A principle of working for an organisation is to comply with the rules.

In order to comply with the rules, you need to know what they are, understand what they mean, and how they relate to you and your role in the organisation.

Problem Statement

In this context, there are a number of challenges for organisations, including

1. how to know all relevant legal and regulatory obligations;
2. how to know which policies and procedures relate to which legal and regulatory obligation;
3. how to know if policies and procedures adequately comply with all legal and regulatory obligations;
4. how to ensure staff understand policies and procedures within which context they must operate;
5. how to ensure operations are performed in line with its prescribed policies and procedures.

Some of these challenges have been discussed recently in a related paper by this student as part of this course (2).

This paper focuses on the general societal problem for all organisations worldwide, of ensuring staff understand its policies and procedures, and considers how the problem can benefit from a Machine Learning-based solution.

Problem Importance

All aspects of an organisation (its technology, data, applications, processes, structures, products, ...) are designed, implemented and maintained by its staff - the people working in the organisation. It follows, therefore, that for staff to follow the rules, they must understand them.

The consequence of staff not understanding the internal rules can be of material importance. Non-compliance with legal or regulatory obligations can result in severe financial and non-financial sanctions, potentially impacting the ongoing viability of the organisation. Consumer trust and brand loyalty can be impaired if there are publicised issues with an organisation's products, services or operations. Performance targets and organisational objectives may not be met.

2 Suitability of ML as a solution

Problem Decomposition

In order to understand the suitability of Machine Learning as a solution, it is helpful to consider the components of the problem (Refer Figure 1).

Figure 1 shows that staff have knowledge gaps in their understanding of policies and procedures. Training is an effective way to improve knowledge, but training may not be sufficiently detailed, or may be out of date, and staff may still have questions that need to be answered. They can ask a colleague who has expertise in the question, but again they are relying on the expert's knowledge to be right. Alternatively they need to find the answer themselves.

The steps to do this *while the question remains unanswered* are

1. locate the right document(s) that relate to the question;
2. look within the document(s) for relevant section(s);
3. read, understand and interpret the guidance in the document(s);
4. validate if this answers the question.

If the question persists then this process continues to iterate until the member of staff finds what they are looking for, or gives up the search.

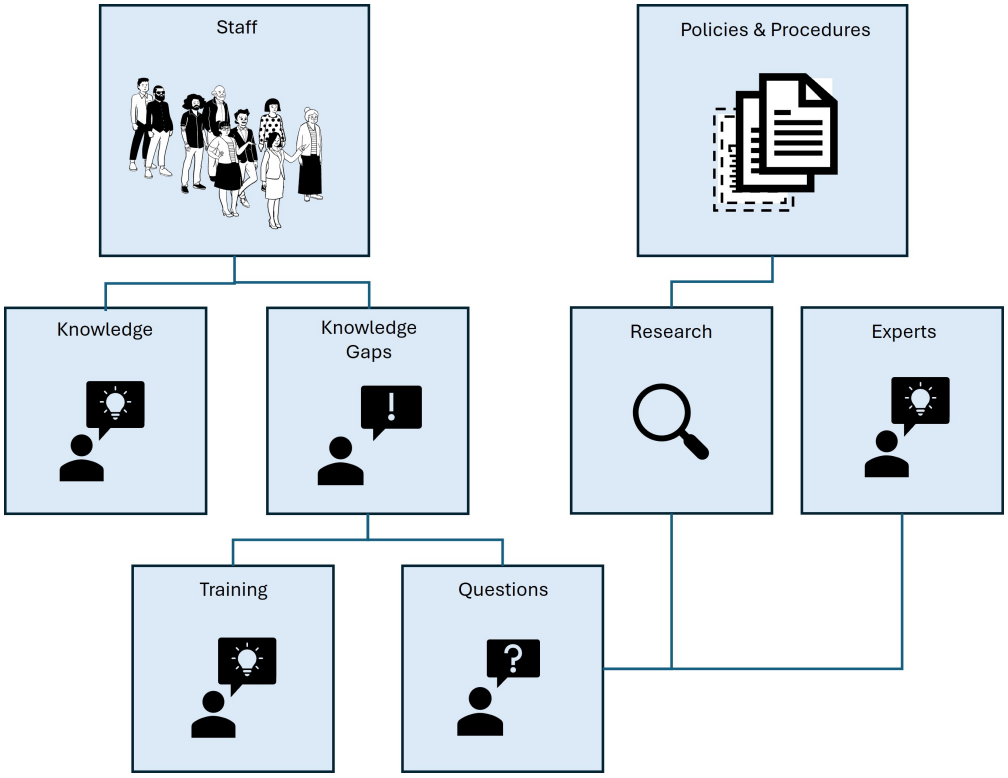

Figure 1: Components of the Problem

This is a time-consuming activity for all staff that have questions, with no certainty on the reliability of results, and hence this manual approach does not provide assurance that questions are answered effectively and accurately.

3 Data Collection & Description

Types of Machine Learning for different data

There are different applications of Machine Learning and different types of Machine Learning are suited to different applications.

A key question is to consider what data is associated with the problem. Data comes in two main categories - structured (typically tables of records/instances and attributes/features) or unstructured.

In this case the primary corpus is a set of policy and procedure documents, stored digitally and accessible to staff. Document text is unstructured. There is no standardisation as to how the documents are organised, or how the text contents.

The next key question to consider, is what type of machine learning approach could be

applied. Supervised learning requires a sample structured training data where each instance has been encoded with one or more labels that shows the desired value to be generated through the machine learning algorithm. The test data has lots of examples so that the machine learning model can detect patterns of features that result in the corresponding label(s). In this way the model can be used to classify or predict the label for new instances of data for which there are only features and no label.

Unsupervised learning is where there are no pre-defined labels on which a model can be trained. There are only features which the model uses to detect patterns to identify similarities such as clustering (looking for associations between different data), or dimensionality reduction (looking to reduce the number of features to those that are most relevant, and discounting features that are less useful in solving the problem)(3).

In this problem there is no structured data, no set of records that can be encoded with labels for training purposes, and hence requires an unsupervised learning approach (3).

This problem requires a contextual *understanding* of text contained in documents. It also requires a similar understanding of a question asked by a person. It then requires a method of searching the documents contents for relevant content that can answer the question, and then provide a response to the question that can be understood by the person, and signpost where the information was found (which document, section, page, etc). This is an appropriate use-case for a machine learning solution based on an application of Document Retrieval and Natural Language Processing (NLP) (4)(5)(6).

Potential Data Sources

An organisation's documents are typically classified according to a Document Policy that determines such attributes as to whether they are confidential, who is permitted to access and read them, where they are stored, how long they should be retained, how often they should be reviewed.

Internal documents are not normally publicly available as a data source. They are normally considered private to the organisation. Each document is normally classified as to who within the organisation can read them, and under what circumstances people outside the oragnisation may read them.

Internal policies and procedures that are published internally to set the standards for staff and operations, should be of good quality. Information should be clear and unambiguous, although this cannot be assumed to be the case.

Sometimes an organisation's policies and procedures assume a level of domain knowledge in the way they are written. Terminology may not always be explained, and acronyms may be used without being defined, where it is considered by an author to be a level of basic assumed knowledge of a reader.

Documents may be written based on a logical decomposition of chapters, sections and sub-sections, although these may not be standardised. Tables of contexts and index are very helpful to readers to identify specific sections of relevance without having to read the entire document to find what is required.

An organisation may have all of their internal documents classified in a library such as an intranet or staff handbook, or alternatively there may be no clear inventory of all documents and where they are stored, or the most recent versions.

Sometimes document repositories have multiple versions of the same document, as it has evolved over time, and publication dates, or version numbers, are required to know which is the most recent and up to date, and discard previous versions as not being an accurate data source.

There is a reliance on staff to ensure that published documents are made available.

There can be a large number of internal documents, and each document can be of many pages of text.

As an example, an organisation in the irish financial sector was reviewed (and remains anonymous for confidentiality) and has a section of its intranet for storing policy and procedure documents. That organisation was found to have 1,172 separate documents, although its not clear if these were all current. Its not clear if this is a typical number of policies and procedures for an organisation, although the assumption is that this is higher than would be typical (its a highly regulated organisation).

The policies are organised by the department responsible for them - for example there are a sub-set of documents that are the responsibility of the Human Resources department.

If each document has on average 30 pages, and if each page contains an average of 15 sentences, that's 527,400 sentences of data to be processed. Assuming each page has on average 200 words, that's equivalent to more than 3 million words of data in total.

Document Metadata

In addition to the content of each document, there are some metadata relating to each document that may be an important consideration, examples include:

- Document ID (Integer)
- Parent Document ID (Integer)
- Document Owner E-mail (String)
- Department ID (Integer) - related to organisation hierarchy table
- Function ID (Integer) - related to organisation hierarchy table

- Team ID (Integer) - related to organisation hierarchy table
- Document Name (String)
- Document Date (Datetime)
- Document Status (Categorical String)
- Internal Source Location URL (String)
- Sourced DateTime (DateTime)
- Internal Staging Location URL (String)

Ethical Considerations

Based on a previously submitted paper as part of this course of study (7), a large number of potential ethical concerns on the use of AI were identified, and categorised thematically under a) Unfair Customer Outcomes, b) Data Processing & Management, and c) Legal & Regulatory Obligations.

Of these, the most relevent for this particular use-case have been directly extracted below:

Consideration 01: Automated Relevancy

"Being able to match a question to the underlying document data and identify the most relevant is not an exact science (matching an explicit value with another explicit value) and hence requires materially important model training and evaluation."

Consideration 02: Unethical Business Behaviour

"It could be considered an 'excuse' for staff non-compliance if the solution does not indicate a policy or procedure *obligation or guideline*, when in effect there should be one. On this basis, there is considered a high importance for human oversight of the testing results of the model."

Consideration 03: Bias in Algorithms

"The documents being used as a basis for responding to questions needs to be limited to the organisation's own documents, and not include any other document that could give inaccurate results."

"Similarly, there could be bias in the model determining what text is relevant in responding to the question."

Consideration 04: Data Traceability

"The document source can be stored as metadata, as well as storing the physical document in a staging area, and not relying on the online version."

"There is no assurance that documents sourced at points in time will remain available online. If a more recent version of a document is published, the previous version may no longer be accessible online."

"It could be considered ethically important that physical files of historic versions of documents are available for audit purposes to show how a question could be answered in one way, at one point in time, based on different documents available."

Consideration 05: Data Limitations

"The solution relies on sourcing *all* relevant documents applicable to the organisation. If this is incomplete, then there is potential for non-compliance (unknown-unknowns)."

Consideration 06: Data Accuracy

"There is an assumption that the documents sourced are reliable in terms of data accuracy. This is of the utmost importance. The source documents may need to be validated by a human expert to ensure they are a reliable source."

"The application of standardised methods for documenting internal procedures, and quality control, is an important consideration to ensure accuracy."

Consideration 07: AI Implementation & Governance Frameworks

"The design, build and test methodology of the solution must be fully integrated with the organisation's AI Implementation & Governance framework to ensure its ethical design and adoption, and assurance of adequate testing."

Consideration 08: Data Privacy

"The internal source documents are typically confidential and the data should not be exposed to publicly available large language models."

"There is potential for personal data to be exposed, e.g. name of process owner, name of policy owner, and their e-mail addresses, although this could be anonymised by storing generic data such as person's role, as opposed to person's name or e-mail address."

4 Proposed ML Solution

Model Type and Justification

As noted previously, the proposed ML Solution will be based on a combination of document retrieval, text extraction and encoding and natural language processing. Machine Learning solutions such as this require a number of tools for different stages of the workflow (6).

All machine learning models take an input (input variables x) and determine a way to map that input to one or more outputs (y). That mapping is described as a function (f) of x. Hence machine learning models can be described as y=f(x).

Different machine learning models will have different results for different data, and selecting the best model is key - this is done based on a thorough understanding of the underlying data, and Exploratory Data Analysis is fundamentally important prior to any model selection or training.

There are broadly two types of machine learning models: *parametric* and *non-parametric*.

Parametric machine learning models determine a mathematical equation with fixed parameters for the function of x which is applied to new instances of data to determine the output. For example the linear function y=ax+b is parametric as the two parameters (coefficient a and constant b) are fixed.

Non-parametric models still use parameters, but the number of parameters is not fixed, and they vary based on the training data presented. Non-parametric models are ideally suited, for this reason to problems involving unstructured data. Examples of non-parametric models include 'k-nearest-neighbour', 'decision trees' and 'support vector machines' (8)(9).

Workflow Overview

The overall solution conceptual schema is shown in Figure 2 highlighting seven steps (A to G).

- A: Identify & Source Valid Documents
- B: Document Representation Stage
- C: Input a Question
- D: Question Representation Stage
- E: Rectrieval & Generation Stage
- F: Model Evaluation Stage
- G: Model Optimisation Stage

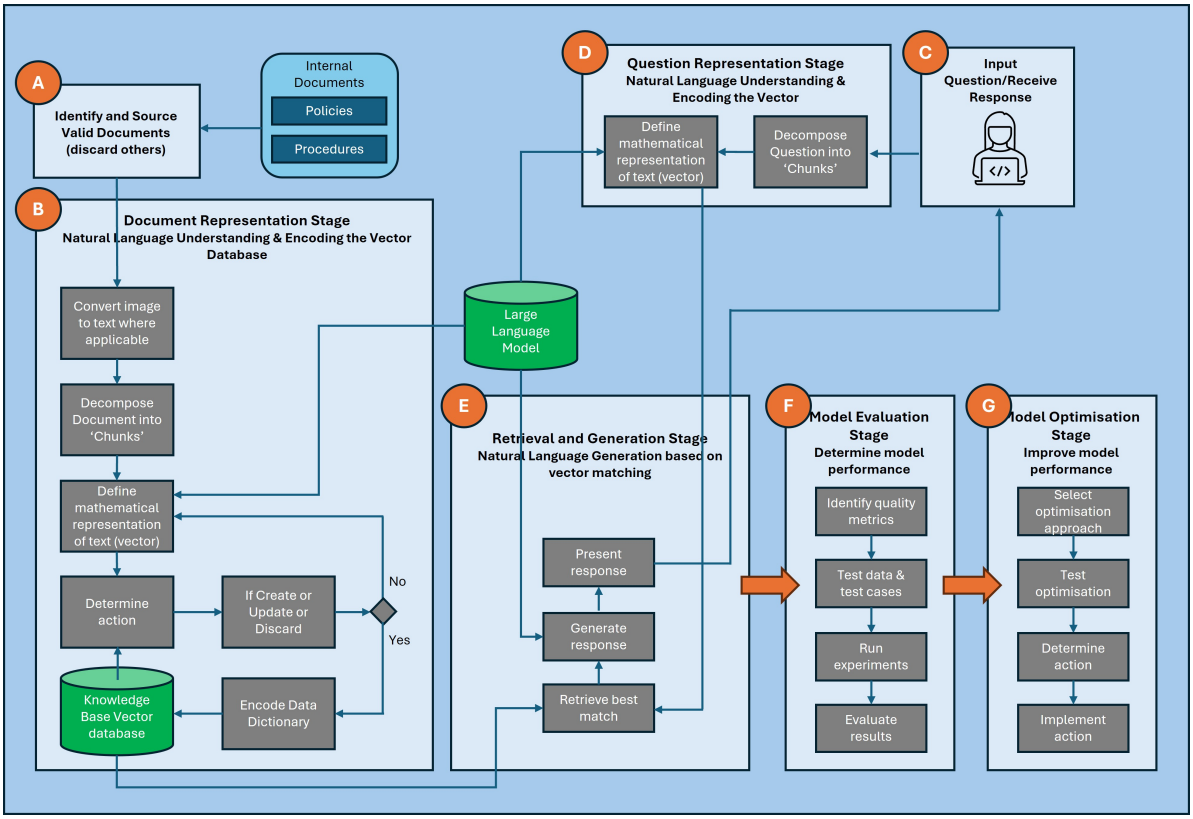

Figure 2: Solution Workflow Overview

A: Identify & Source Valid Documents

A routine that scans file locations and determines if a document is a valid source data. If valid then retrieve the document to a staging area.

B: Document Representation Stage

Decompose the document into small 'chunks' and use a model to represent the chunk as a mathematical vector. Store the vector in the knowledge base database.

C: Input a Question

A user inputs a question (using an input form procedure of the solution application).

D: Question Representation Stage

The question is decomposed into one or more 'chunks' and a model is used to represent the chunk as a mathematical vector.

E: Rectrieval & Generation Stage

The question vector is looked up in the knowledge base vector database and the best matches are determined, and a large language model is used to generate a natural language response to the user.

F: Model Evaluation Stage

The model is *evaluated* to determine performance (accuracy) based on controlled document input and controlled user question. Experiments are performed and results evaluated.

G: Model Optimisation Stage

Various optimisation approaches may be used to improve the model. After each optimisation, the results are evaluated to determine if accuracy has improved. Any improvements are retained, otherwise the optimisation is discarded.

5 Evaluation Metrics and Expected Outcomes

As shown in Figure 2, the Model Evaluation Stage considers the quality metrics that are considered important, along with specific test data and test cases to run experiments in a controlled way, to evaluate results.

As this is a controlled test, each test case has a question prompt, and an 'ideal' response that should be generated by the model. The model's generated response as part of the test is compared with the 'ideal' response to determine a rating for accuracy.

It is possible to automate the evaluation testing also using a large language model to compare the generated response vs. the 'ideal' response, although there should be a level of human oversight to review the evaluations.

A benchmark 'accuracy score' can be developed to act as a threshold for a test case pass/fail.

As the response is a generated natural language response, based on the underlying knowledge base form the internal documents, it is vitally important that sufficient breadth and depth of test cases validate that the model is effective with accurate responses.

6 Challenges and Limitations

There are a number of challenges in ensuring that this Machine Learning-based solution delivers accurate responses to staff questions.

All of the eight ethical issues as previously noted in Section 3 need to be considered as part of the solution design, to mitigate risks.

The solution is applicable to a variety of similar use cases of questions and answers based on a specific document knowledge base. The solution is only as good as that knowledge base. This means a) the underlying data must be complete and accurate, and b) the selected model must successfully decompose and encode the data for contextual retrieval.

As new data sources are refreshed in the knowledge base, it is important that the solution is evaluated and optimised as applicable.

As with all machine-learning applications, there is a level of human oversight required to ensure the model continues to perform effectively.

If a response is deemed inaccurate or nonsense, then there should be a feedback process by which the staff member can flag the question and response so that this can be tested, evaluated and corrected.

With so many organisations world-wide, it should be possible to develop a standard application and implementation approach by which a model can be trained on a particular organisation's documents, and an instance of the application and data staging environment implemented to ensure effective results in a secure, private and safe environment.

Further areas of research can now progress to detailed design and selection of tools for the various workflow components.

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